

SOCIOL 723: Social Statistics II

Week 13, Causal Machine Learning

Wenhao Jiang

Department of Sociology, Fall 2026



Today

Why Naive ML Fails

Double Machine Learning

Heterogeneous Treatment Effects

Applications

Wrap-Up

★ = essential, you must understand this

★ = for understanding, not examined

Putting the Two Halves Together ★

- **Weeks 5–11** gave us identification: what must be true for a parameter to be causal. Almost every design ended with a nuisance function to estimate, a propensity score, an outcome regression, a first stage, a conditional trend.
- **Week 12** gave us flexible estimators for exactly such functions, which work with many covariates and unknown functional form.
- **The obvious move, plug ML into the causal estimator, fails.** Today is about why, and what fixes it.

The payoff

We get to relax the functional-form assumptions that identification arguments have been quietly leaning on all semester, AJR's linear geography control, the linear propensity score, the additive conditional trend, *without* giving up $\sqrt{n}$ inference.

Plan

Goal: valid $\sqrt{n}$ inference on a causal parameter when the nuisance functions are estimated by machine learning, and honest estimates of how effects vary.

1. Why plugging ML into a regression fails: regularization bias, shown in a simulation; single versus double selection.
2. Neyman orthogonality, derived; why the product of errors is what matters.
3. Cross-fitting and the DML recipe; its variance; the ATE score.
4. The 401(k) data: OLS, single and double lasso, DML with three learners, sensitivity to omitted confounders.
5. Heterogeneous effects: meta-learners, causal forests, BLP, GATES, RATE, policy learning, on the 401(k) data.
6. Applications in social science: growth mindset, college and low-wage work.

Scripts: `figs/w13_sims.R`, `figs/w13_401k.R`. Book: Chernozhukov, Hansen, Kallus, Spindler, and Syrgkanis (2024).

The Problem

The Partially Linear Model ★

To see the problem without letting notation hide it, start with the simplest causal case there is: *one* treatment effect to estimate, but completely flexible relationships between the controls, the treatment, and the outcome. That is the **partially linear model**, the workhorse for the rest of the course:

$$\begin{aligned} Y_i &= \theta D_i + g(W_i) + \epsilon_i, & E[\epsilon_i \mid D_i, W_i] &= 0 \\ D_i &= m(W_i) + v_i, & E[v_i \mid W_i] &= 0. \end{aligned}$$

- θ is the target: the effect of D holding W fixed, and it is assumed constant (relaxed in the last section).
- g and m are **nuisance functions**; we do not care about them, but we must handle them. They may be highly nonlinear, and W may be high-dimensional.
- Under conditional ignorability given W (Week 5), θ is the ATE.
- Week 1 solved this when g and m are linear and low-dimensional. The question now is what happens when we estimate them with lasso or a forest.

Attempt 1: Regularization Bias ★

The naive approach: fit the partially linear model by ML to get $\hat{g}$, taking it from a joint fit of Y on (D_i, W_i) with D_i left *unpenalized* so that $\hat{g}$ targets the structural g . Then regress $Y_i - \hat{g}(W_i)$ on D_i .

$$\hat{\theta}_{\text{naive}} = \frac{\frac{1}{n} \sum_i D_i (Y_i - \hat{g}(W_i))}{\frac{1}{n} \sum_i D_i^2}.$$

Substituting the true model and rescaling,

$$\sqrt{n}(\hat{\theta}_{\text{naive}} - \theta) = \underbrace{\left(\frac{1}{n} \sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}} \sum_i D_i \epsilon_i}_{\xrightarrow{d} \mathcal{N}(0, \Sigma)} + \underbrace{\left(\frac{1}{n} \sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}} \sum_i D_i (g(W_i) - \hat{g}(W_i))}_{\text{regularization bias}}.$$

- Regularized estimators are *deliberately biased* (Week 12), and typically converge more slowly than the parametric $n^{-1/2}$.

Attempt 1: Regularization Bias ★ (cont.)

- Then the second term does not vanish on the $\sqrt{n}$ scale, the nuisance bias is of the same order as, or larger than, the sampling noise we are trying to describe.
- The point estimate may be fine; the inference is not. Confidence intervals are centred in the wrong place, and collecting more data does not fix it.
- **A trap worth naming.** If you instead regress Y on W alone you estimate $\ell(W) = E[Y | W] = \theta m(W) + g(W)$, which already contains the effect you are after. Subtracting that removes the signal. And even with the true ℓ , regressing $Y - \ell(W)$ on *raw* D_i does not recover θ : D_i must be residualized too, which is exactly what the next slide does.

Attempt 2: Partial Out Both Sides ★

Frisch–Waugh–Lovell says: residualize Y and D on W , then regress. With ML estimates $\hat{g}$ and $\hat{m}$,

$$\tilde{Y}_i = Y_i - \hat{\ell}(W_i), \quad \tilde{D}_i = D_i - \hat{m}(W_i), \quad \hat{\theta} = \frac{\frac{1}{n} \sum_i \tilde{D}_i \tilde{Y}_i}{\frac{1}{n} \sum_i \tilde{D}_i^2},$$

where $\ell(W) = E[Y \mid W]$.

The bias term now involves the **product** of the two estimation errors:

$$\text{bias} \approx \sqrt{n} \cdot \frac{1}{n} \sum_i (m(W_i) - \hat{m}(W_i)) (\ell(W_i) - \hat{\ell}(W_i)).$$

This vanishes when $\|\hat{m} - m\|_2 \|\hat{\ell} - \ell\|_2 = o_p(n^{-1/2})$, for which it is sufficient that *each* converge faster than $n^{-1/4}$. Lasso and forests can do that under sparsity or smoothness assumptions. **First-order bias has become second-order.**

Seeing the Bias: a Simulation

$Y = 0.5 D + g(W) + \epsilon$, $D = m(W) + v$; random forests for the nuisances; 300 draws (figs/w13_sims.R).

estimator	A: smooth ($n = 500$)		B: very nonlinear ($n = 1,000$)	
	bias / sd	coverage	bias / sd	coverage
OLS on (D, W)	-0.08		+11.2	
naive plug-in ($Y - \hat{g}$ on raw D)	-3.20		+6.40	
partialling out, in-sample forests	+0.11	0.94	+3.57	0.04
DML, 5-fold cross-fitting	+0.28	0.93	+3.96	0.01
RMSE of $\hat{m}$, $\hat{\ell}$	0.41, 0.35		0.43, 1.02	

- **A** (Chernozhukov et al.'s Figure 1 design): the naive plug-in is off by three sds; partialling out is centred and covers.
- **B**: forests cannot learn g here (RMSE of $\hat{\ell}$ 1.02); DML is biased too. **Orthogonality makes errors multiply; it does not make bad estimates good.**
- In-sample forests matched cross-fitting here; learners that overfit more do not.

This Is Neyman Orthogonality Again ★

The residualized moment condition is

$$E[\psi(W_i; \theta, \eta)] = 0, \quad \psi = [(Y - \ell(W)) - \theta(D - m(W))] (D - m(W)),$$

and it satisfies

$$\left. \frac{\partial}{\partial \eta} E[\psi(W_i; \theta_0, \eta)] \right|_{\eta=\eta_0} = 0.$$

- **You met this property in Week 6:** the AIPW estimator's bias was the *product* of the two nuisance errors, so first-order mistakes in either one cancel. The partially linear case is the same statement for a different moment.
- And you derived it in **Week 1**: FWL is the linear special case, in which $\hat{m}$ is an OLS projection and the “adaptivity” of $\hat{\beta}_1$ to estimation error in the residuals is exactly orthogonality.
- **The thread of the whole course:** FWL $\rightarrow$ AIPW $\rightarrow$ DML. One idea, three levels of generality.

Orthogonality, Derived

Perturb the nuisances in some direction: $\ell_r = \ell_0 + r \delta_\ell$, $m_r = m_0 + r \delta_m$. With $V = D - m_0(W)$ and $E[V | W] = 0$:

$$\psi_r = [Y - \ell_r - \theta_0(D - m_r)](D - m_r) \quad \text{the score}$$

$$\begin{aligned} \frac{\partial}{\partial r} E[\psi_r] \Big|_{r=0} &= E[(-\delta_\ell + \theta_0 \delta_m) V] - E[(Y - \ell_0 - \theta_0 V) \delta_m] \quad \text{product rule} \\ &= 0 - E[\epsilon \delta_m] = 0 \quad E[V | W] = 0, E[\epsilon | W] = 0 \end{aligned}$$

Compare the naive score $\varphi = (Y - \theta D - g(W)) D$:

$$\frac{\partial}{\partial r} E[(Y - \theta_0 D - g_0 - r \delta_g) D] \Big|_{r=0} = -E[\delta_g(W) D] = -E[\delta_g(W) m_0(W)] \neq 0.$$

A small error in $\hat{g}$ moves the naive moment at first order; it moves the orthogonal one only at second order, through products of errors.

Where the Product Comes From

Write $V = D - m(W)$, $\epsilon = Y - \ell(W) - \theta_0 V$, and the nuisance errors $\Delta_m = \hat{m} - m$, $\Delta_\ell = \hat{\ell} - \ell$. The cross-fitted score at θ_0 is

$$(Y_i - \hat{\ell}_i - \theta_0(D_i - \hat{m}_i))(D_i - \hat{m}_i) = (\epsilon_i - \Delta_{\ell i} + \theta_0 \Delta_{mi})(V_i - \Delta_{mi}).$$

Multiply out and average:

$$\frac{1}{n} \sum_i \psi_i = \underbrace{\frac{1}{n} \sum \epsilon_i V_i}_{\text{CLT term}} - \underbrace{\frac{1}{n} \sum (\Delta_{\ell i} - \theta_0 \Delta_{mi}) V_i}_{\text{mean zero}} - \underbrace{\frac{1}{n} \sum \epsilon_i \Delta_{mi} + \frac{1}{n} \sum (\Delta_{\ell i} - \theta_0 \Delta_{mi}) \Delta_{mi}}_{\text{product of errors}}$$

- The middle terms have mean zero because Δ was estimated on *other* folds: it is fixed given those folds, while V_i and ϵ_i have mean zero. That is what cross-fitting buys.
- The last term is of order $\|\Delta_m\| \|\Delta_\ell\| + \|\Delta_m\|^2$; it must be $o(n^{-1/2})$, which holds if each error is $o(n^{-1/4})$.

Single Selection Versus Double Selection

Single selection: lasso of Y on W picks the controls; then OLS of Y on D and those controls.

- A confounder that predicts D strongly but Y only moderately is likely dropped: its coefficient in the Y -lasso is small, and D is not there to stand in for it. Omitted-variable bias returns.

Double selection: union of the controls chosen to predict Y *and* those chosen to predict D ; then OLS. A variable important for either equation is kept: the lasso version of partialling out both sides.

abortion and crime, 48 states $\times$ 12 years	violent	property	murder
Donohue and Levitt (2001)	-0.129 (0.024)	-0.091 (0.018)	-0.121 (0.047)
all 251 controls	0.294 (0.472)	-0.068 (0.157)	0.321 (1.109)
double selection	-0.087 (0.181)	-0.094 (0.051)	0.006 (0.280)

Double selection keeps initial abortion rates and their trends, which predict abortion with $R^2 \approx 0.9$; the violent-crime and murder effects lose significance.

DML

The Second Ingredient: Cross-Fitting ★

Orthogonality is not quite enough. If $\hat{m}$ is fitted on the *same* observations where it is evaluated, its errors correlate with those units' outcomes, **overfitting bias**, which orthogonality does not remove.

Cross-fitting

Split the sample into K folds. For each fold k : estimate $\hat{\ell}^{(-k)}, \hat{m}^{(-k)}$ on the *other* folds, and form residuals only for observations *in* fold k . Then pool.

$$\hat{\theta}_{\text{DML}} = \frac{\frac{1}{n} \sum_k \sum_{i \in \mathcal{I}_k} (D_i - \hat{m}^{(-k)}(W_i)) (Y_i - \hat{\ell}^{(-k)}(W_i))}{\frac{1}{n} \sum_k \sum_{i \in \mathcal{I}_k} (D_i - \hat{m}^{(-k)}(W_i))^2}$$

This is the same device as cross-validation in Week 12, but used to protect *inference*, not to select a tuning parameter.

The DML Recipe ★

1. Split the sample into K folds (typically $K = 5$; split on the *independent unit* if data are clustered).
2. On the training folds, estimate $\hat{\ell}(W) = \hat{E}[Y | W]$ and $\hat{m}(W) = \hat{E}[D | W]$ with any ML method, lasso, random forest, boosting, or an ensemble.
3. On the held-out fold, form residuals $\tilde{Y}_i$ and $\tilde{D}_i$.
4. Pool across folds and regress $\tilde{Y}$ on $\tilde{D}$.
5. With independent observations, the standard error is the usual heteroskedasticity-robust one from that final bivariate regression: **orthogonality is what makes it valid** despite the ML first stage. *With clustered data, keeping clusters intact within folds is not enough*, the score variance must also be clustered at the independent unit.

The DML Recipe ★ (cont.)

6. Repeat the whole split several times with different seeds and take the median; the answer should not depend on the partition.

Chernozhukov et al. (2018). In R: DoubleML with `mlr3` learners.

The DML Variance

Write the score as $\psi = (\tilde{Y} - \theta \tilde{D})\tilde{D}$ with $\tilde{Y} = Y - \ell(W)$, $\tilde{D} = D - m(W)$. A linear score gives a ratio, and a ratio gives a sandwich:

$$\hat{\theta} - \theta_0 = \frac{\frac{1}{n} \sum_i (\tilde{Y}_i - \theta_0 \tilde{D}_i) \tilde{D}_i}{\frac{1}{n} \sum_i \tilde{D}_i^2} \quad \text{subtract } \theta_0 \text{ from the ratio}$$

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0, \frac{E[\psi^2]}{J^2}\right), \quad J = E[\tilde{D}^2] \quad \text{CLT on top; LLN below}$$

- The estimate of $E[\psi^2]/J^2$ is the heteroskedasticity-robust variance of the final residual-on-residual regression (Week 2), computed with cross-fitted residuals.
- Orthogonality is what lets the estimated $\hat{\ell}, \hat{m}$ stand in for the true ones without adding a term to this variance.
- With S repeated splits: take the median estimate, and inflate the variance by the spread across splits.

The ATE Score: AIPW Is DML

Binary D , no constant-effect assumption: target the ATE with outcome models $m_d(x) = E[Y \mid D = d, X = x]$ and propensity $e(x)$ (Week 6):

$$\psi_i = m_1(X_i) - m_0(X_i) + \frac{D_i(Y_i - m_1(X_i))}{e(X_i)} - \frac{(1 - D_i)(Y_i - m_0(X_i))}{1 - e(X_i)} - \theta.$$

- Orthogonal in (m_0, m_1, e) : the error is $E[(\hat{e} - e)(\hat{m}_1 - m_1)/\hat{e}]$ plus the analogous term for m_0 , a product.
- With cross-fitted ML for the three functions, this is DML for the **interactive regression model**. It estimates the ATE even when effects vary; the partially linear model estimates a variance-weighted average of them.
- Weights $1/\hat{e}$ explode near 0 or 1: check overlap, trim, or target the ATT or the overlap-weighted effect (Week 6).

DML for Instruments: the Partially Linear IV Model

$Y = \theta D + g(W) + \epsilon$ with $E[\epsilon | Z, W] = 0$; D endogenous, Z an instrument valid only conditional on W (Week 7):

$$\psi = [(Y - \ell(W)) - \theta(D - r(W))](Z - q(W)), \quad \hat{\theta} = \frac{\sum_i (Z_i - \hat{q})(Y_i - \hat{\ell})}{\sum_i (Z_i - \hat{q})(D_i - \hat{r})}$$

with ℓ, r, q the conditional means of Y, D, Z given W , cross-fitted.

- A Wald ratio of residuals: partial W out of Y, D , and Z by ML, then IV.
- Card's proximity instrument (Week 7) is credible only given region, urbanicity, and family background; this lets them enter flexibly.
- Weak-instrument problems survive: a small $\sum (Z - \hat{q})(D - \hat{r})$ is still a small denominator; Anderson–Rubin logic still applies.

Folds, Clusters, and Panels

- Cross-fitting assumes the held-out fold is independent of the training folds. With students in schools or person-years in a panel, **split by school or person**; otherwise the same person's other years leak into the nuisance fit and the overfitting bias returns.
- The variance must also be clustered at that level (sum the score within cluster before squaring).
- **Repeated splits**: the estimate depends on the random partition; repeat S times, report the median, and add the between-split spread to the variance. The 401(k) estimates use 5 folds $\times$ 5 repetitions.
- **How many folds**: 5 is typical; 2 wastes data in each nuisance fit. DML2 (pool the score across folds) is preferred over averaging fold-specific estimates.

What the Theory Requires, and What It Does Not ★

Requires

- Conditional ignorability given W (unchanged!).
- Overlap: $\hat{m}(W)$ bounded away from 0 and 1 for binary D .
- Nuisance convergence fast enough that the *product* of the two rates is $o(n^{-1/2})$; each faster than $n^{-1/4}$ is sufficient but not necessary. Approximate sparsity for lasso, smoothness for forests.
- Orthogonal moment + cross-fitting.

Does not require

- Correct functional form for g or m .
- Knowing which of the p covariates matter.
- $p \ll n$. High-dimensional W is the point.
- Consistent *model selection*, lasso may pick the wrong variables and $\hat{\theta}$ is still fine.

What the Theory Requires, and What It Does Not ★ (cont.)

The assumption that does not go away

DML does not identify anything. It is an estimation technology for a parameter already identified by the arguments of Weeks 5–11. If W omits a confounder, no amount of machine learning helps.

Running Example: 401(k) Eligibility and Wealth

The question. Does being eligible for a 401(k) plan raise household net financial assets? Eligibility is set by the employer, so it is not randomly assigned: it is plausibly ignorable only after conditioning on income, age, and other characteristics that predict both job choice and saving (Poterba, Venti, and Wise 1994; Chernozhukov et al. 2018).

The data. SIPP 1991, 9,915 households (`hdm::pension`). 37% eligible. Controls: age, income, education, family size, married, two earners, defined-benefit pension, IRA, home owner.

Why ML: the confounders enter through income and age in ways no one would write down; a linear model in nine controls is a strong functional-form claim.

Plan: raw gap; OLS; single vs. double lasso on a 139-column dictionary; DML, partially linear and interactive, with lasso, forests, and boosting.

401(k): One Effect, Many Estimators

Effect of eligibility on net financial assets, dollars (robust se) (figs/w13_401k.R):

estimator	nuisances	estimate	se
difference in means	none	19,559	1,413
OLS, 9 linear controls	linear	5,896	1,524
single selection (lasso of Y only)	9 of 139 terms	8,099	
double selection	14 of 139 terms	8,956	1,290
DML, partially linear	lasso (139 terms)	9,609	1,343
	random forest	8,965	1,306
	boosting	9,907	1,368
DML, interactive (ATE)	lasso	9,319	3,065
	random forest	8,022	1,214
	boosting	8,232	1,239

Flexible adjustment lands near \$8,000–10,000 whatever the learner; Chernozhukov et al. (2018) report 9,247 and 8,105 with forests. The partially linear model weights effects by $\text{Var}(D | X)$; the interactive one targets the ATE (lasso's se is large: propensities near 0.04).

How Robust? Sensitivity for DML

Chernozhukov, Cinelli, Newey, Sharma, and Syrgkanis (forthcoming) extend Week 6's omitted-variable sensitivity to DML: how strong would an unobserved confounder have to be?

- Bound the bias by two partial R^2 's: how much of the residual variation in the outcome, and in the treatment (its Riesz representer), the confounder explains, C_Y^2 and C_D^2 .
- 401(k), DML with forests: partially linear estimate 9,002 (se 1,394); **robustness value** 5.4%: a confounder explaining 5.4% of both residual variations would bring the estimate to zero.
- A confounder as strong as $C_Y^2 \approx 4\%$, $C_D^2 \approx 3\%$ (their benchmark, calibrated on observed variables): bias at most 4,196; bounds [4,808; 13,196]. Still positive.

Same logic as Cinelli and Hazlett (Week 6), with the treatment equation replaced by the Riesz representer, which covers the ATE score as well as the partially linear one.

Where DML Plugs In

The same two ingredients, an orthogonal score plus cross-fitting, apply to almost every design in this course:

Design	Nuisance functions	Week
Partially linear regression	$E[Y W], E[D W]$	1
Interactive / ATE under CIA (AIPW)	$E[Y D, W], \Pr(D = 1 W)$	6
Instrumental variables	$E[Y W], E[D W], E[Z W]$	7
Regression discontinuity	$E[Y W]$ within the bandwidth	9
Conditional DID	outcome trend and propensity, by cohort	10

- For IV: residualize Y , D , and Z on W using ML, then run 2SLS on the residuals. This is what lets us replace AJR's "linear in distance from the equator plus continent dummies" with something flexible.

Where DML Plugs In (cont.)

- For conditional DID: this is precisely the doubly robust Callaway–Sant’Anna estimator from Week 10, with ML nuisances.

Recent Directions

- **Automatic debiasing** (Chernozhukov, Newey, and Singh 2022): every orthogonal score in this lecture is “plug-in + a weight $\times$ residual”. The weight (the Riesz representer: $1/e(X)$ terms for the ATE) can be learned directly by lasso, without deriving its formula, for any linear functional (average derivatives, policy effects).
- **Deep learning for causal effects**: networks with a shared representation of X and one output head per treatment arm, sometimes with a third head for the propensity; train, then plug the heads into the AIPW score (Koch et al. 2025, a primer for sociologists).
- **Text and images as confounders**: embeddings (Week 12) enter W ; the orthogonal score makes small errors in those representations second order.
- **Software**: DoubleML (R, Python), `grf`, and `hdm`; the textbook is Chernozhukov, Hansen, Kallus, Spindler, and Syrgkanis (2024).

HTE

From ATE to CATE ★

Drop the constant-effect assumption. The target becomes the **conditional average treatment effect**

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x].$$

- Sociologically this is often the substantive question: *for whom* does college pay off, *which* neighbourhoods benefit, *who* is harmed.
- The naive approach, run subgroup regressions, report the significant ones, is a **multiple-testing disaster** and does not replicate. It is the specification search of Week 2 with more rooms to search.
- The modern approach: build an orthogonal, doubly robust *pseudo-outcome* whose conditional mean is $\tau(x)$, then use ML to estimate that conditional mean, with honest sample splitting so that the subgroups are not chosen using the same data that tests them.

Meta-Learners: Five Ways to Estimate $\tau(x)$

With $\mu_d(x) = E[Y \mid D = d, X = x]$ and $e(x) = \Pr(D = 1 \mid X = x)$:

learner	recipe	weakness
S	one model $\hat{\mu}(x, d)$; $\hat{\tau}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$	regularization can shrink d out
T	separate $\hat{\mu}_1, \hat{\mu}_0$; subtract	errors of two fits add
X	impute each unit's effect from the <i>other</i> group's model, then smooth, weighted by $e(x)$ (Künzel et al. 2019)	good when groups differ in size
R	minimize $\sum_i [(Y_i - \hat{\ell}(X_i)) - (D_i - \hat{e}(X_i))\tau(X_i)]^2$ (Nie and Wager 2021)	needs orthogonal residuals
DR	regress the AIPW pseudo-outcome Γ_i on X_i (Kennedy 2023)	extreme $1/\hat{e}$

- R and DR are *orthogonal*: their error beyond an oracle is a product, $(\hat{e} - e)(\hat{\mu}_d - \mu_d)$, as in DML.
- The R-learner is Robinson's partialling out with θ replaced by $\tau(X)$.

The Doubly Robust Score as a Pseudo-Outcome ★

Take the AIPW summand from Week 6, unit by unit:

$$\Gamma_i = \hat{m}_1(X_i) - \hat{m}_0(X_i) + \frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{m}_0(X_i))}{1 - \hat{e}(X_i)}.$$

- Its *average* is the AIPW estimate of the ATE; that was Week 6.
- With the *true* nuisance functions, its conditional mean is exactly the CATE: $E[\Gamma_i \mid X_i = x] = \tau(x)$. With cross-fitted estimates it holds approximately, at a rate governed by the same product condition as before.
- So: construct Γ_i with cross-fitted nuisances, then **regress Γ_i on X_i by any method you like**.
 - Regress on a few chosen covariates → **group average treatment effects** (GATEs) with valid standard errors.
 - Regress with a forest → a nonparametric CATE surface.
 - Regress on a constant → back to the ATE.
- **This one construction unifies the whole section.**

Causal Forests ★

Wager and Athey (2018), Athey, Tibshirani, and Wager (2019): a random forest whose splits are chosen to maximize *heterogeneity in the treatment effect* rather than in the outcome.

- **Honesty:** within each tree, one subsample chooses the splits and a *disjoint* subsample estimates the effects in the resulting leaves. This is what delivers asymptotic normality and valid pointwise intervals, without it, leaves are selected *because* they show large effects.
- **Local centering:** residualize Y and D on X first (i.e. orthogonalize), exactly as in DML.
- **Output:** an estimate $\hat{\tau}(x)$ for every unit, with standard errors, plus variable-importance measures for *effect* heterogeneity.
- `grf::causal_forest()`.

How a Causal Forest Estimates $\tau(x)$

A forest defines **weights**: $\alpha_i(x)$ = how often unit i shares a leaf with the point x , averaged over trees. Then solve a local, residualized regression:

$$\hat{\tau}(x) = \frac{\sum_i \alpha_i(x) (D_i - \hat{e}(X_i))(Y_i - \hat{\ell}(X_i))}{\sum_i \alpha_i(x) (D_i - \hat{e}(X_i))^2}$$

- The partially linear (Robinson) estimator, but weighted toward units near x : DML made local. With $\alpha_i(x) = 1/n$ it is the DML estimate.
- Splits are chosen to make $\hat{\tau}$ differ most between children (a gradient approximation keeps it fast) (Athey, Tibshirani, and Wager 2019).
- **Honesty**: half of each tree's subsample chooses the splits, the other half fills the leaves. **Subsampling** (not bootstrap) gives the variance for pointwise intervals (Wager and Athey 2018).

Reporting Heterogeneity Honestly ★

An estimated $\hat{\tau}(x)$ that varies is not by itself evidence of heterogeneity, noise varies too.

- **Calibration test:** regress the doubly robust scores on $\hat{\tau}(X_i)$ (out of fold). A slope near 1 indicates real, well-calibrated heterogeneity; a slope near 0 says your CATE surface is noise. `grf::test_calibration()`.
- **Sorted group ATEs:** rank units by $\hat{\tau}$, form quantile groups, and report the doubly robust ATE *within each group* using held-out data. Then test whether the top and bottom groups differ.
- **Pre-specify** the moderators you care about wherever possible, and label exploratory heterogeneity as exploratory.
- **Effect heterogeneity is where causal ML is most useful and most abused.** The honest-splitting discipline is not optional.

Testing for Heterogeneity: BLP and GATES (Chernozhukov, Demirer, Duflo, and Fernández-Val 2025)

Given a proxy $S(X)$ for the CATE, built on one half of the data, use the other half:

BLP: $\Gamma_i = \beta_1 + \beta_2(S(X_i) - \bar{S}) + \text{error}$ β_1 : ATE; $\beta_2 = 1$ if S is the true CATE

GATES: $\gamma_k = E[\Gamma_i \mid S(X_i) \in \text{quintile } k]$ effects in groups sorted by S

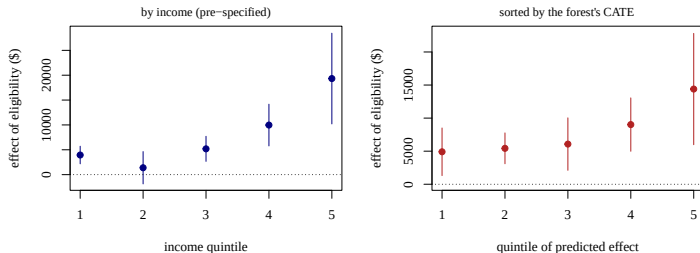
- $\beta_2 = 0$ means no detectable heterogeneity along S ; the test is valid even if S is a bad estimate (a bad S just has low power).
- **CLAN**: compare covariate means between the most and least affected groups.
- Repeat over many splits and report medians (with adjusted p -values).

India immunization incentives (25 vs. 78 villages): ATE 2.8 shots per village per month; most-affected fifth +13.2, least-affected -8.0, significantly negative.

Heterogeneity in the 401(k) Data

Causal forest on the nine controls; doubly robust scores for every comparison

(figs/w13_401k.R):



- ATE 7,964 (1,124). Calibration: mean prediction 1.10 (good); differential prediction 0.03 ($p = 0.47$): the forest's own ranking carries no detectable signal.
- By pre-specified income quintile: 3,946 (904) at the bottom to 19,326 (4,656) at the top. Partly scale: dollars of saving rise with income.
- RATE on a held-out half: ranking by the forest 1,253 (1,928); by income 9,661 (5,974). Targeting gains are not established.

Should We Target? Rank-Weighted ATEs (Yadlowsky et al. 2025)

A prioritization rule $S(X)$ ranks units; treat the top share u first.

$$\text{TOC}(u) = E[\tau(X) \mid S(X) \text{ in the top } u] - \text{ATE} \quad \text{gain from targeting the top } u$$

$$\text{RATE} = \int_0^1 \alpha(u) \text{TOC}(u) du \quad \text{AUTOOC: } \alpha = 1; \text{ Qini: } \alpha(u) = u$$

- $\text{RATE} > 0$: the rule finds units with larger effects. Estimate S on one sample, RATE on another.
- Works for any rule: a forest's $\hat{\tau}(x)$, a risk score, or income.

Policy learning (Athey and Wager 2021): choose a treatment rule $\pi(x) \in \{0, 1\}$ from a simple class (a depth-2 tree) to maximize $\frac{1}{n} \sum_i (2\pi(X_i) - 1) \Gamma_i$; regret shrinks like $\sqrt{\text{complexity}/n}$. The DR scores again do the work.

Intervals for Individual Effects (Lei and Candès 2021)

$\hat{\tau}(x)$ with a standard error describes the *average* effect among people like x . A person's own effect $Y_i(1) - Y_i(0)$ varies around it.

- Weighted split conformal (Week 12): build a prediction interval for the missing $Y_i(1)$ or $Y_i(0)$, reweighting calibration cases by the propensity to correct for covariate shift between groups.
- Combine the two intervals into an interval for the individual effect, with coverage guaranteed if the propensity is known (randomized experiments), approximately otherwise.
- Typical finding: individual intervals are wide, often covering zero, even when the CATE is precisely estimated. Statements about “who benefits” are about groups.

Applications

In the Literature: Growth Mindset in Schools (Athey and Wager 2019)

The question. A short online intervention teaches students that intelligence can grow. For whom does it raise achievement?

The data. **Simulated** from a model fit to the National Study of Learning Mindsets (a randomized study) by the organizers of a 2018 workshop: 10,391 students in 76 schools. The simulated data show selection, so the authors treat them as observational. A demonstration of method, not a finding about mindsets.

The design. A causal forest with propensity and outcome forests for local centering, clustered by school; heterogeneity checked by the calibration test and by pre-specified school-level moderators.

AIPW average effect	0.247 (± 0.04)
above- minus below-median predicted effect	0.053 (± 0.071)
calibration: differential prediction	0.322 (se 0.307, $p = 0.29$)

Athey and Wager: What the Forest Could and Could Not Say

- The overall effect is clear. The forest's own ranking of students does not separate large from small effects: the omnibus test finds no heterogeneity it can trust.
- One **pre-specified** school characteristic, the school's average fixed mindset, shows credible moderation in a test on school-level DR scores ($t = -3.02$, $p = 0.004$); the authors add that it is unclear whether it survives controls for other school covariates.
- The forest split on that variable often (24% of depth-weighted splits), but splits are not tests: the test came from the DR scores.
- Lesson for practice: a CATE surface that varies is not evidence of heterogeneity; report the calibration test, and keep a short list of moderators chosen before looking.

In the Literature: Brand, Xu, Koch, and Geraldo (2021)

The question. Who benefits most from college, in avoiding low-wage work? Theories of positive selection (those likeliest to go gain most) and negative selection (the disadvantaged gain most) predict opposite patterns.

The data. NLSY79, ages 14–17 in 1979, completed at least 12th grade: 4,548 respondents. Outcome: share of 1990–2014 spent in low-wage work (below two thirds of the median hourly wage).

The design. *Causal trees* with honest splitting: one half of the sample grows the tree, the other estimates the effect in each leaf, with propensity-score matching inside leaves; sensitivity analysis for unobserved confounding.

Average effect: college reduces time in low-wage work by about 18 percentage points (matching), 17 (causal forest).

Brand et al.: Who Gains Most

The tree's first split is mother's education:

leaf	effect of college (pts)
mother without a high-school degree	−26 pts
and a large family	−33 pts
mother high school or more, low income	−19 pts
father did not attend college	−21 pts
all respondents	−17 to −18 pts

- Larger effects for the more disadvantaged: **negative selection**, and a case for expanding access at the margin where college is least likely.
- The leaves are found by the algorithm, not by the authors' hypotheses; honest estimation keeps the effects in each leaf from being selected because they are large.

Numbers from the 2019 working version; see also Brand, Zhou, and Xie (2023) for a review.

Wrap-Up

Where This Leaves the Course

- **One idea, three appearances.** FWL (Week 1) $\rightarrow$ double robustness and Neyman orthogonality (Week 6) $\rightarrow$ DML and causal forests (today). Each generalizes the last; none replaces identification.
- **ML relaxes functional-form assumptions, not identifying assumptions.** That is a real and substantial gain, many published identification arguments lean on linearity they never defended, but it is a bounded one.
- **The binding constraint is still the design.** A causal forest on a confounded comparison estimates heterogeneity in a bias.

Next week

Synthesis, and a review for the final exam.

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